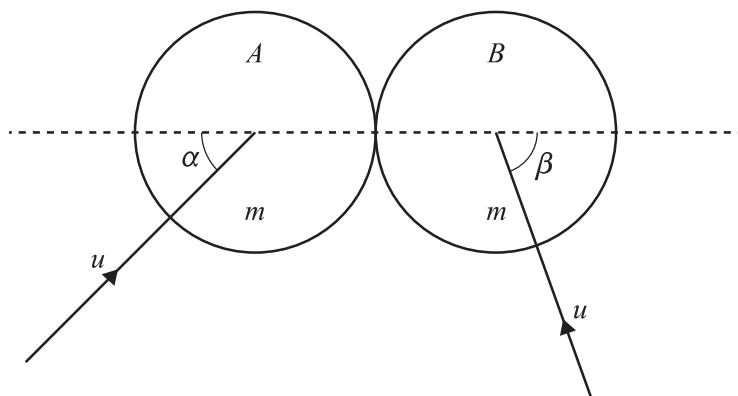




(a) Show that the direction of motion of  $A$  after the collision is perpendicular to the line of centres. [4]

This image shows a full page of a worksheet designed for handwriting practice. It features approximately 20 horizontal dashed lines spaced evenly across the page, providing a guide for letter height and placement. The background is plain white, and there are no other markings or text present.

The total kinetic energy of the spheres after the collision is  $\frac{3}{4}mu^2$ .

(b) Find the value of  $\alpha$ .

[4]

[illegible]